

Southeast University

Department of Computer Science and Engineering

PROJECT PROPOSAL

Asset Management System

A Software-Based Solution for Managing
Organizational Assets

Course: Advanced Java Lab

Course Code: CSE352.4

Submitted By

No.	Student Name	Student ID
1.	Md. Mehedy Hasan	2022100010064
2.	Md. Jakir Hossain	2022100010065
3.	Ahnaf Sakil Mahmud	2022100010066

Submitted To

Miftahul Sheikh

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1. Introduction / Abstraction

Asset management is an important part of maintaining the operational efficiency, financial accountability, and security of an organization. Organizations may own hundreds or thousands of assets such as computers, mobile devices, office equipment, vehicles, furniture, networking devices, and other resources.

Managing these assets manually using spreadsheets or paper-based records can lead to duplicate records, inaccurate information, loss of assets, difficulty in tracking ownership, and inefficient maintenance processes.

The proposed **Asset Management System** is a centralized software solution designed to digitize and simplify the complete asset lifecycle. The system will allow an organization to register assets, assign them to employees or departments, track their locations and status, maintain asset history, record maintenance activities, and monitor asset availability.

The application will use **Flutter** as the frontend technology, providing a modern cross-platform user interface. The backend will be developed using **Spring Boot**, which will expose secure RESTful APIs and handle business logic, authentication, authorization, and asset-related operations.

The system will use **PostgreSQL** as its relational database for reliable and structured data storage.

The overall objective is to provide a reliable, transparent, and scalable platform through which organizations can maintain accurate asset records and make data-driven decisions regarding asset utilization, maintenance, replacement, and retirement.

Technology Stack

Layer	Technology
Frontend	Flutter / Dart
Backend	Spring Boot / Java
API Communication	REST API / JSON
Database	PostgreSQL
Authentication	JWT-based Authentication
Architecture	Client-Server Architecture

Table 1: Proposed Technology Stack

2. Feature

The Asset Management System will provide the following major features.

2.1 User Authentication and Authorization

The system will provide secure login and role-based access control. Different users will have different permissions according to their responsibilities.

Possible roles include:

- System Administrator
- Asset Manager
- Department Manager
- Employee / Asset User

Role-based authorization will prevent unauthorized users from modifying sensitive asset information.

2.2 Asset Registration

Administrators or authorized asset managers will be able to register new assets with information such as:

- Asset name and category
- Unique asset identification number
- Serial number
- Purchase date and purchase price
- Vendor or supplier
- Warranty information
- Current location
- Asset condition and status

2.3 Asset Assignment

An asset can be assigned to an employee, department, branch, or other responsible entity.

The system will maintain the assignment date and assignment history so that the organization can determine who was responsible for an asset at a particular point in time.

2.4 Asset Tracking

Users will be able to search and filter assets based on:

- Category
- Status
- Location
- Department
- Assigned employee
- Asset identification number

The system will provide a clear view of the current state and ownership of every asset.

2.5 Maintenance Management

Maintenance records can be associated with assets.

Users can record maintenance requests, service dates, maintenance costs, service providers, and descriptions of performed work.

This will help organizations monitor recurring maintenance and estimate future maintenance requirements.

2.6 Asset Transfer

Authorized users will be able to transfer assets between employees, departments, or locations.

Each transfer will be stored as part of the asset's historical record.

2.7 Asset Lifecycle Management

The system will support different stages of an asset's lifecycle:

Purchased → Available → Assigned → Maintenance → Returned → Retired

This makes it easier to understand how assets move through the organization from acquisition to retirement.

2.8 Dashboard and Reports

A dashboard will provide summarized information such as:

- Total number of assets
- Available assets
- Assigned assets
- Assets under maintenance
- Retired or disposed assets
- Asset distribution by department or location

Reports can help management understand asset utilization and identify areas where assets are underutilized or require attention.

3. Unique Feature

The proposed system will differentiate itself through a combination of **complete asset lifecycle tracking, asset history, smart identification, and accountability features.**

3.1 Complete Asset History

Instead of storing only the current state of an asset, the system will preserve important events throughout its lifecycle.

For example, an asset's history can show when it was purchased, who received it, when it was transferred, when it entered maintenance, and when it was retired.

This historical timeline can provide a strong accountability mechanism and help organizations investigate missing or damaged assets.

3.2 QR Code / Barcode Based Identification

Each asset can be assigned a unique QR code or barcode.

By scanning the code using the Flutter application, an authorized user can quickly retrieve the asset's information without manually entering an identification number.

The scan can provide:

- Asset details

- Current assigned user
- Current location
- Asset condition
- Maintenance history
- Assignment and transfer history

3.3 Asset Accountability

When an asset is assigned to a person or department, the system records the responsible party and assignment history.

This creates a clear accountability trail and reduces disputes regarding asset ownership or responsibility.

3.4 Smart Notifications

The system can generate notifications for events such as:

- Upcoming warranty expiration
- Scheduled maintenance
- Overdue asset returns
- Pending asset transfer approvals
- Assets remaining in maintenance for an extended period

3.5 Utilization-Based Decision Support

By analyzing asset assignment, status, and historical records, the system can help management identify unused or underutilized assets.

This can reduce unnecessary purchases and improve resource allocation.

4. UML / ERD Diagram

The following diagram represents the main entities and relationships of the proposed Asset Management System.

4.1 Entity Relationship Diagram

The Asset Management System is designed around an organization or company that owns and manages its assets. An organization can have multiple departments, users, locations, categories, and assets.

An asset can be assigned to a user through an assignment record. Assignments are stored separately so that the system can maintain the complete history of asset assignments and transfers.

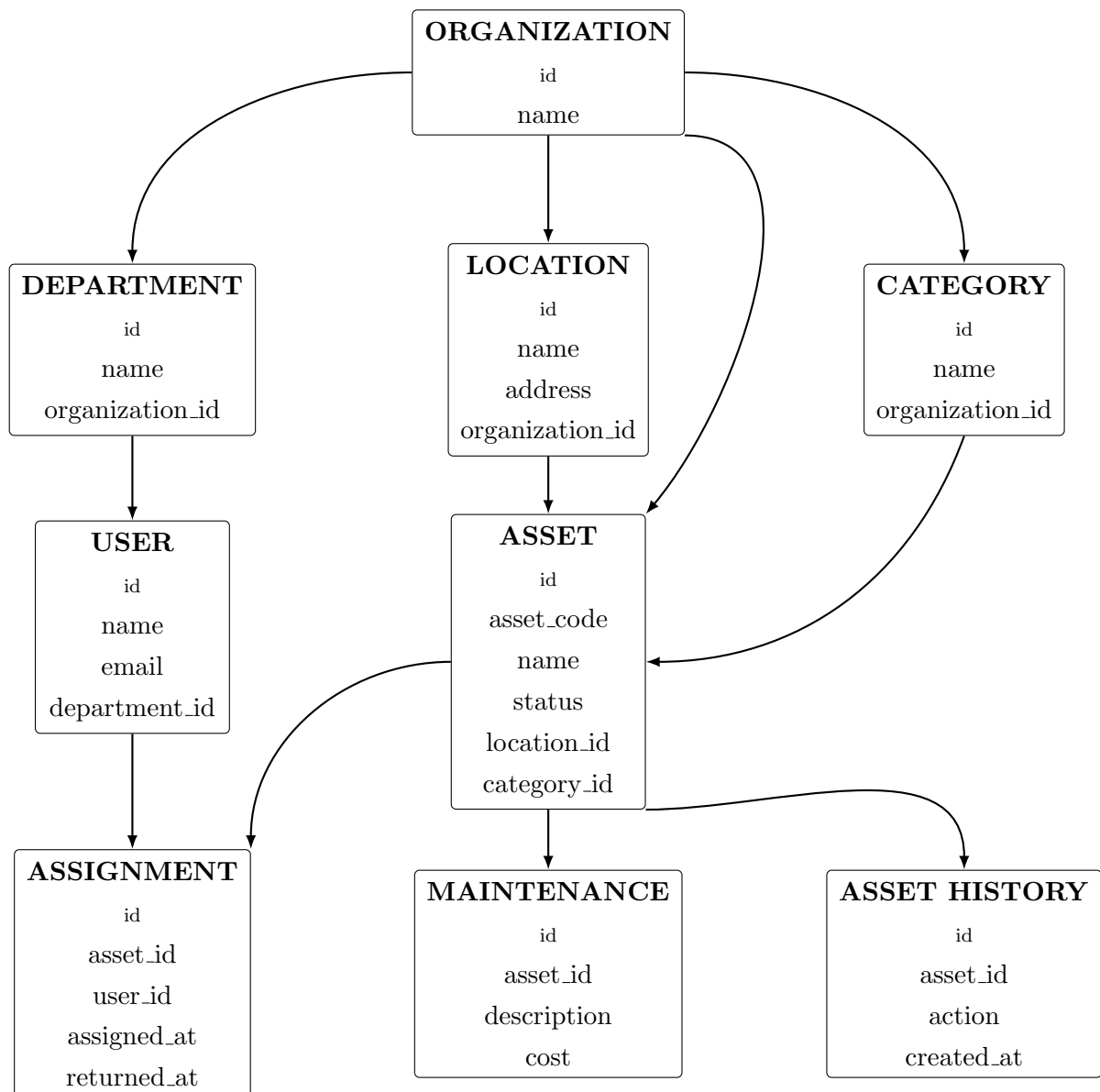


Figure 1: Entity Relationship Diagram of the Asset Management System

4.2 High-Level System Architecture

The system will follow a client–server architecture.

The Flutter application will communicate with the Spring Boot backend through REST APIs. The backend will validate requests, apply business rules, perform authentication and authorization, and communicate with PostgreSQL.

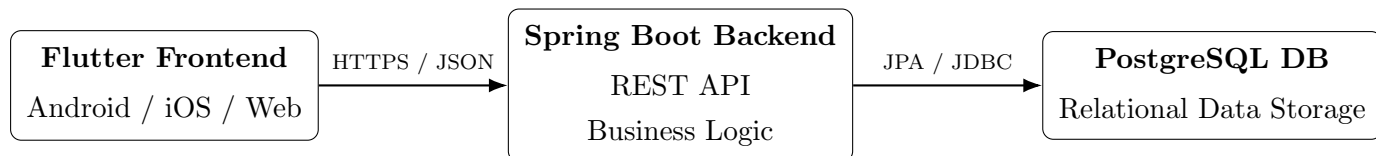


Figure 2: High-Level System Architecture

5. Social Impact

A digital asset management system can have a positive impact on organizations by improving transparency, accountability, and resource efficiency.

5.1 Reduced Resource Waste

Accurate records can prevent organizations from purchasing equipment that is already available but difficult to locate.

Identifying unused assets can also allow resources to be reassigned instead of purchasing new ones.

5.2 Improved Accountability

Every assignment and transfer can be recorded digitally.

This makes it easier to identify the responsible person or department and can reduce the risk of lost or improperly managed assets.

5.3 Improved Operational Efficiency

Replacing spreadsheets and manual paperwork with a centralized system reduces administrative workload.

Employees can quickly search for asset information, while management can access summarized information through dashboards and reports.

5.4 Better Maintenance and Sustainability

Tracking maintenance helps organizations keep equipment operational for longer.

Extending the useful life of devices and equipment can reduce unnecessary electronic waste and support more sustainable resource consumption.

5.5 Data-Driven Management

Management can use asset-related data to make better decisions regarding purchasing, replacement, maintenance, and resource distribution.

This promotes more efficient use of organizational budgets.

6. Conclusion

The proposed Asset Management System is intended to provide a centralized, secure, and efficient solution for managing organizational assets throughout their complete lifecycle.

By combining Flutter, Spring Boot, and PostgreSQL, the project will provide a modern cross-platform interface, a scalable backend, and a reliable relational data layer.

The system will address common problems associated with manual asset tracking, including inaccurate records, inefficient searching, unclear ownership, unrecorded transfers, and poor maintenance visibility.

Features such as QR or barcode identification, complete asset history, accountability tracking, and smart notifications will further improve the usefulness of the platform.

Ultimately, the project aims to help organizations use their assets more efficiently, reduce unnecessary costs, improve accountability, and make better decisions based on reliable asset data.

The architecture will also allow the system to be extended in the future with additional modules such as advanced analytics, mobile push notifications, audit logs, integrations with other enterprise systems, and automated reporting.